№3 18.09.2024 No-3

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# Аппроксимация частных производных

$$x_i = x_0 + ih \qquad y_j = y_0 + jh \qquad \bar X_{(i)} = B_h \cdot (i) + \bar X_0$$

решётка Браве

I в $R^2$ Формулы естественной аппроксимации производных

$$\int_\Gamma P\,dx + Q\,dy = \iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy \quad 2D$$

$$\Downarrow$$

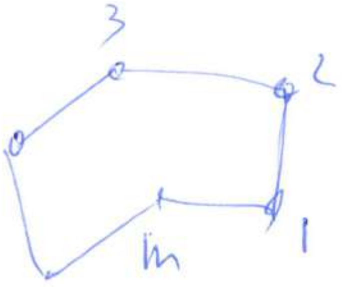
$$\int_\Gamma f\,dx = -\iint_S \frac{\partial f}{\partial y}\,dx\,dy = -\frac{\partial f}{\partial y}\Big|_{(x_0,y_0)} \cdot S$$

$$\int_\Gamma f\,dy = \iint_S \frac{\partial f}{\partial x}\,dx\,dy = \frac{\partial f}{\partial x}\Big|_{(x_0,y_0)} \cdot S$$

$$\Rightarrow \frac{\partial f}{\partial x} = \frac{\int_\Gamma f\,dy}{S} \qquad \frac{\partial f}{\partial y} = \frac{-\int_\Gamma f\,dx}{S}$$

$$S = \iint_S 1\,dx\,dy = \int_\Gamma x\,dy = -\int_\Gamma y\,dx$$

$$\Rightarrow \frac{\partial f}{\partial x} = \frac{\int_\Gamma f\,dy}{\int_\Gamma x\,dy} \qquad \frac{\partial f}{\partial y} = \frac{\int_\Gamma f\,dx}{\int_\Gamma y\,dx}$$



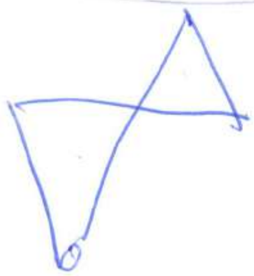

$$2\int_\Gamma f\,dx = \sum_{i=1}^{m} f_i(\cancel{x}y_{i+1} - \cancel{x}y_{i-1})$$

$$2\int_\Gamma y\,dx = \sum_{i=1}^{m} \cancel{x}y_i(\cancel{x}y_{i+1} - \cancel{x}y_{i-1})$$

$m+1 \to 1$

$0 \to m$

$$\boxed{\frac{\partial f}{\partial x} \approx \frac{\sum_{i=1}^{m} f_i(y_{i+1}-y_{i-1})}{\sum_{i=1}^{m} x_i(y_{i+1}-y_{i-1})} \qquad \frac{\partial f}{\partial y} \approx \frac{\sum_{i=1}^{m} f_i(x_{i+1}-x_{i-1})}{\sum_{i=1}^{m} y_i(x_{i+1}-x_{i-1})}}$$



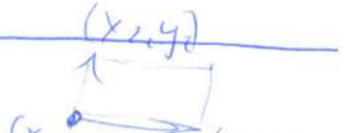
$(x, y) \to (x^1, x^2)$

$$\boxed{\frac{\partial f}{\partial x^j} = \frac{\sum_{i=1}^{m} f_i(x^{3-j}_{i+1} - x^{3-j}_{i+1})}{\sum_{i=1}^{m} x^j(x^{3-j}_{i+1} - x^{3-j}_{i-1})}}$$

$\int_{S'} \bar{a}\cdot\bar{n}\,dS' = \int_V \operatorname{div}\bar{a}\,dV$ Гаусса-Остроградского 3D

2D

$n=2,\ m=3$ – Треугольник




$$2V = \begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix} = \begin{vmatrix} 0 & x_1-x_3 & y_1-y_3 \\ 0 & x_2-x_3 & y_2-y_3 \\ 1 & x_3 & y_3 \end{vmatrix} = \begin{vmatrix} x_1-x_3 & y_1-y_3 \\ x_2-x_3 & y_2-y_3 \end{vmatrix} \qquad \frac{\partial f}{\partial x} = \frac{V_1}{V}$$

$$2V_1 = \begin{vmatrix} 1 & f_1 & y_1 \\ 1 & f_2 & y_2 \\ 1 & f_3 & y_3 \end{vmatrix} = \begin{vmatrix} f_1-f_3 & y_1-y_3 \\ f_2-f_3 & y_2-y_3 \end{vmatrix}, \quad 2V_2 = \begin{vmatrix} 1 & x_1 & f_1 \\ 1 & x_2 & f_2 \\ 1 & x_3 & f_3 \end{vmatrix} = \begin{vmatrix} x_1-x_3 & f_1-f_3 \\ x_2-x_3 & f_2-f_3 \end{vmatrix} \quad \frac{\partial f}{\partial y} = \frac{V_2}{V}$$

3D $\searrow$ $\boxed{3}$

$n=3,\ m=4$ Тетраэдр

$$6V = \begin{vmatrix} 1 & x_1 & y_1 & z_1 \\ 1 & x_2 & y_2 & z_2 \\ 1 & x_3 & y_3 & z_3 \\ 1 & x_4 & y_4 & z_4 \end{vmatrix} = \begin{vmatrix} x_1-x_4 & y_1-y_4 & z_1-z_4 \\ x_2-x_4 & y_2-y_4 & z_2-z_4 \\ x_3-x_4 & y_3-y_4 & z_3-z_4 \end{vmatrix}$$

$$6V_1 = \begin{vmatrix} 1 & f_1 & y_1 & z_1 \\ 1 & f_2 & y_2 & z_2 \\ 1 & f_3 & y_3 & z_3 \\ 1 & f_4 & y_4 & z_4 \end{vmatrix} = \begin{vmatrix} f_1-f_4 & y_1-y_4 & z_1-z_4 \\ f_2-f_4 & y_2-y_4 & z_2-z_4 \\ f_3-f_4 & y_3-y_4 & z_3-z_4 \end{vmatrix} \Rightarrow V_1 = \frac{1}{6}\begin{vmatrix} f_1-f_4 & y_1-y_4 & z_1- \\ f_2-f_4 & y_2-y_4 & z_2- \\ f_3-f_4 & y_3-y_4 & z_3- \end{vmatrix}$$

$$\frac{\partial f}{\partial x} = \frac{V_1}{V},\quad \frac{\partial f}{\partial y} = \frac{V_2}{V},\quad \frac{\partial f}{\partial z} = \frac{V_3}{V}$$

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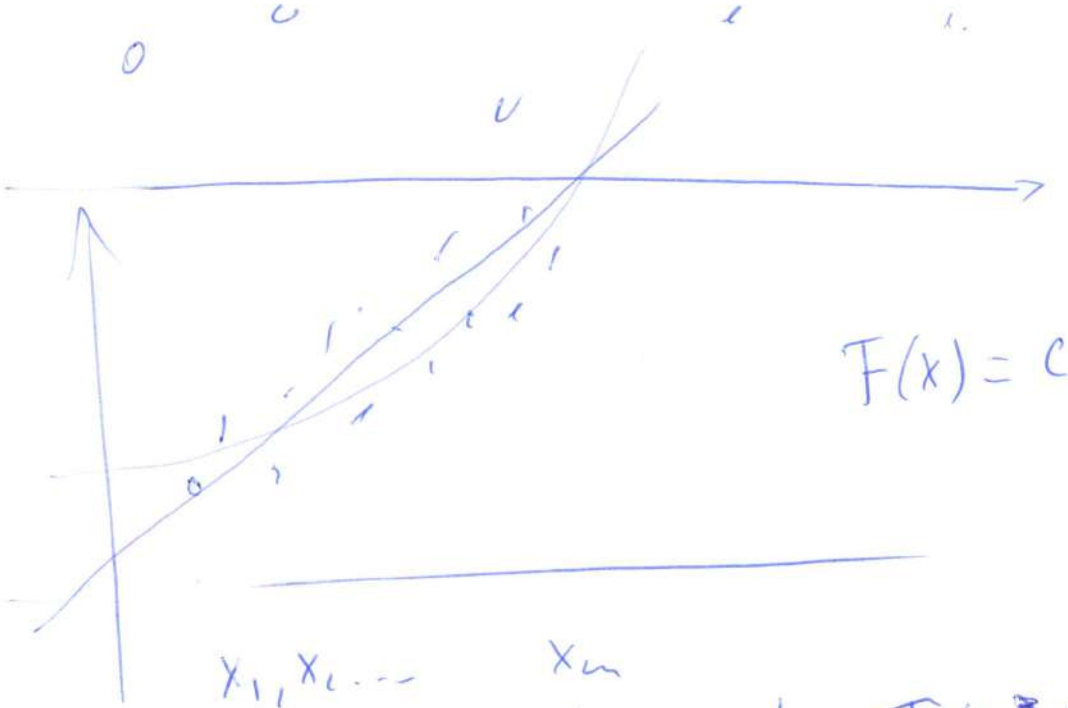
$n$-~~мбсе~~ любое, $m=n+1 \longrightarrow nD$

$$n!\,V = \begin{vmatrix} 1 & x_1^1 & \dots & x_1^n \\ \vdots & & & \\ 1 & x_m^1 & \dots & x_m^n \end{vmatrix} = \begin{vmatrix} x_1^1-x_m^1 & & x_1^n-x_m^n \\ & \dots & \\ x_n^1-x_m^1 & & x_n^n-x_m^n \end{vmatrix}$$

$$n!\,V_i = \dots \quad \Rightarrow \quad \boxed{\frac{\partial f}{\partial x^i} = \frac{V_i}{V}}$$

метод наименьших квад.

# Метод наименьших квадратов



$$F(x) = c_1 + c_2 x$$

$x_1, x_2 \dots \quad x_m$

$f_1, f_2 \quad\quad f_m$

$c_j$ - параметры

$$f \sim F(x, c_1, c_2, \dots, c_n)$$

$n \leq m$

$$\boxed{S = \sum_{i=1}^{m} \left[ f_i - F(x_i, c_1 \dots c_n) \right]^2}$$

сумма квадратов разности

коэфф.

Найти коэффициенты $c_1, c_2, \dots c_n$, такие, что

достигает мин.

$S$ достигает минимума. $\Rightarrow$

$$\Rightarrow \left\{ \frac{\partial S}{\partial c_j} = 0 \right. \qquad \frac{\partial S}{\partial c_j} = \sum_{i=1}^{m} 2\left[ f_i - F(x_i, c_1 \dots c_n) \right]\left( -\frac{\partial F}{\partial c_j} \right) = 0$$

мет. наим. квд.

МНК для аппроксимации частных производных

размерность пространства

$n$ - размерность пространства

$m$ - число точек (узлов) $(m > n)$

Пусть $\bar{x}_1, \bar{x}_2, \dots \bar{x}_m \in R^n$

$f_1, f_2 \dots f_m$

nD linear function

$$F(x^1, \dots, x^n) = A_0 + A_1 x^1 + A_2 x^2 + \dots + A_n x^n$$

$A_0$ - значение функции в точке → знач. функц. в точке

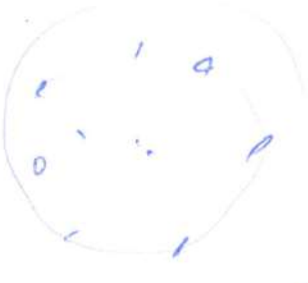
$x_c$ центральная точка

$$A_i = \frac{\partial F}{\partial x^i} \qquad \bar{x}_c = \frac{1}{m}\sum_{i=1}^{m} \bar{x}_i$$

$$\bar{F}(x^1 \dots x^n) = A_0 + A_1(x^1 - x^1_c) + \dots + A_n(x^n - x^n_c)$$

$$\begin{cases} \dfrac{\partial S}{\partial A_0} = 0 \\ \dfrac{\partial S}{\partial A_1} = 0 \\ \vdots \\ \dfrac{\partial S}{\partial A_n} = 0 \end{cases} \qquad \boxed{S = \sum_{i=1}^{m} (F(\bar{x}_i) - f_i)^2 \to} \min$$

квадр. отклонение



$A_0, A_1, \dots, A_n$ пара...

$A_0, A_1, \dots, A_n$

→ СЛАУ относительно $A_0, A_1, \dots, A_n$

Формула Крамера

$$\left|\begin{matrix} a_{11} & \dots & a_{1n} \\ \vdots & & \\ a_{n1} & & a_{nn} \end{matrix}\right|\begin{matrix} b_1 \\ \\ b_n \end{matrix} \qquad \Delta = \left|\begin{matrix} a_{11} & \\ & a_{nn} \end{matrix}\right|, \ \Delta_1 = \left|\begin{matrix} b_1 & a_{12} & a_{1n} \\ \vdots & & \\ b_n & a_{n2} & a_{nn} \end{matrix}\right|, \ \Delta_2$$

$$X_i = \frac{\Delta_i}{\Delta}$$

Решение $A_0 = \frac{1}{m}\sum_{i=1}^{m} f_i$

$V = \{v_{ij}\} = \{x_i^j - x_m^j\}$

$$V = \begin{array}{c} \begin{array}{ccc} x_1 \ldots x_n & y_1 & y_m \\ \downarrow \quad \downarrow & \downarrow & \downarrow \end{array} \\ \begin{bmatrix} x_1^1 - x_m^1 & x_1^2 - x_m^2 & x_1^n - x_m^n \\ x_2^1 - x_m^1 & x_2^2 - x_m^2 & x_2^n - x_m^n \\ \vdots & & \\ x_{m-1}^1 - x_m^1 & x_{m-1}^2 - x_m^2 & x_{n-1}^n - x_m^n \end{bmatrix} \end{array} \; m.$$

$V_{(m-1)\times n}$

$V_i$ ~~получен~~ получаем из $V$ ~~заменой~~ заменой $i$-го ~~столбца~~ столбца на столбец
$$\begin{pmatrix} f_1 - f_m \\ f_2 - f_m \\ \\ f_{m-1} - f_m \end{pmatrix}$$

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$A_0 = \frac{1}{m}\sum_{i=1}^{m} f_i$

$$A_j = \frac{\partial f}{\partial x_j} = \frac{\det|V^T \cdot V_j|}{\det|V^T \cdot V|}$$ ← формула Крамера

~~не если~~ но если $m = n+1 \Rightarrow A_j = \dfrac{\det V_j}{\det V}$

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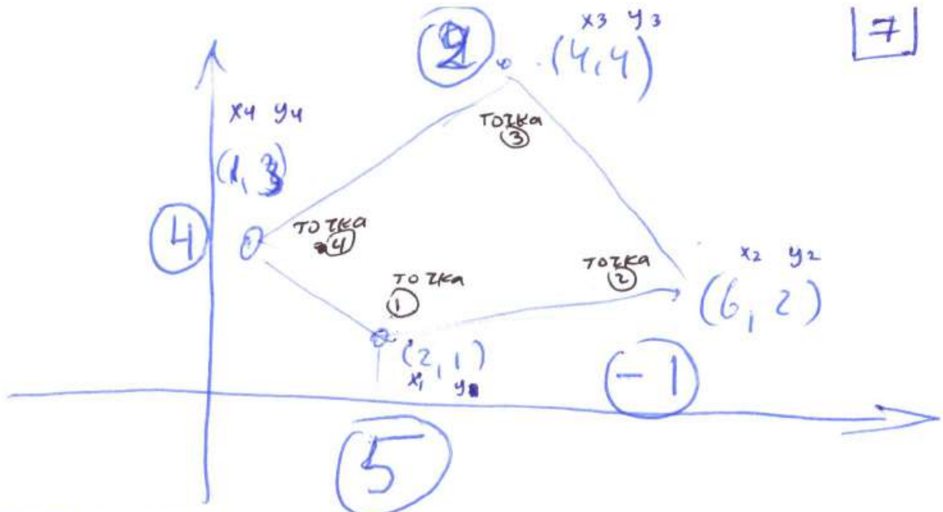
$$\boxed{\begin{aligned} A_0 &= f = \frac{1}{m}\sum_{i=1}^{m} f_i \\ A_j &= \frac{\det|V^T V_j|}{\det|V^T \cdot V|} \end{aligned}}$$




Естественная апроксимация производных:

$$\frac{\partial f}{\partial x} \approx \frac{\sum_{i=1}^{4} f_i (y_{i+1} - y_{i-1})}{\sum_{i=1}^{4} x_i (y_{i+1} - y_{i-1})} =$$

$$= \frac{f_1(y_2 - \overset{y_4}{y_0}) + f_2(y_3 - y_1) + f_3(y_4 - y_2) + f_4(\overset{y_1}{y_5} - y_3)}{x_1(y_2 - y_0) + x_2(y_3 - y_1) + x_3(y_4 - y_2) + x_4(y_5 - y_3)}$$

$$= \frac{5(2-3) + (-1)(4-1) + 2(3-2) + 4(1-4)}{2(2-3) + 6(4-1) + 4(3-2) + 1(1-4)}$$

$$= \frac{-5 - 3 + 2 - 12}{-2 + 18 + 4 - 3} = \frac{-18}{17} \approx -1.0588$$

$$\frac{\partial f}{\partial y} \approx \frac{\sum_{i=1}^{4} f_i (x_{i+1} - x_{i-1})}{\sum_{i=1}^{4} y_i (x_{i+1} - x_{i-1})}$$

$$= \frac{f_1(x_2 - x_0) + f_2(x_3 - x_1) + f_3(x_4 - x_2) + f_4(x_5 - x_3)}{y_1(x_2 - x_0) + y_2(x_3 - x_1) + y_3(x_4 - x_2) + y_4(x_5 - x_3)}$$

$$= \frac{5(6-1) + (-1)(4-2) + (2)(1-6) + 4(2-4)}{(1)(6-1) + (2)(4-2) + (4)(1-6) + (3)(2-4)}$$

$$= \frac{25 - 2 - 10 - 8}{5 + 4 - 20 - 6}$$

$$= \frac{5}{-17} \approx -0.29412$$

МНК: метод наименьших квадр.

$$V = \begin{bmatrix} x_1 - x_4 & y_1 - y_4 \\ x_2 - x_4 & y_2 - y_4 \\ x_3 - x_4 & y_3 - y_4 \end{bmatrix}_{3\times 2} = \begin{bmatrix} 2-1 & 1-3 \\ 6-1 & 2-3 \\ 4-1 & 4-3 \end{bmatrix}_{3\times 2} = \begin{bmatrix} 1 & -2 \\ 5 & -1 \\ 3 & 1 \end{bmatrix}_{3\times 2}$$

$$\begin{bmatrix} f_1 - f_4 \\ f_2 - f_4 \\ f_3 - f_4 \end{bmatrix} = \begin{bmatrix} 5-4 \\ -1-4 \\ 2-4 \end{bmatrix} = \begin{bmatrix} 1 \\ -5 \\ -2 \end{bmatrix}_{3\times 1}, \quad V_1 = \begin{bmatrix} 1 & -2 \\ -5 & -1 \\ -2 & 1 \end{bmatrix}_{3\times 2}, \quad V_2 = \begin{bmatrix} 1 & 1 \\ 5 & -5 \\ 3 & -2 \end{bmatrix}_{3\times 2}$$

$$V^T V_1 = \begin{bmatrix} 1 & 5 & 3 \\ -2 & -1 & 1 \end{bmatrix}_{2\times 3} \begin{bmatrix} 1 & -2 \\ -5 & -1 \\ -2 & 1 \end{bmatrix}_{3\times 2} = \begin{bmatrix} -30 & -4 \\ 1 & 6 \end{bmatrix} \Rightarrow \det(V^T V_1) = -176$$

$$V^T V_2 = \begin{bmatrix} 1 & 5 & 3 \\ -2 & -1 & 1 \end{bmatrix}_{2\times 3} \begin{bmatrix} 1 & 1 \\ 5 & -5 \\ 3 & -2 \end{bmatrix}_{3\times 2} = \begin{bmatrix} 25 & -30 \\ -4 & 1 \end{bmatrix} \Rightarrow \det(V^T V_2) = -85$$

$$V^T V = \begin{bmatrix} 1 & 5 & 3 \\ -2 & -1 & 1 \end{bmatrix}_{2\times 3} \begin{bmatrix} 1 & -2 \\ 5 & -1 \\ 3 & 1 \end{bmatrix}_{3\times 2} = \begin{bmatrix} 35 & -4 \\ -4 & 6 \end{bmatrix} \Rightarrow \det(V^T V) = 194$$

$$\frac{\partial f}{\partial x} \approx \frac{\det|V^T V_1|}{\det|V^T V|} = \frac{-176}{194} \approx -0.9072$$

$$\frac{\partial f}{\partial y} \approx \frac{\det|V^T V_1|}{\det|V^T V|} = \frac{-85}{194} \approx -0.438$$

неправильно

неправильная формула.